Vishnu N. Athreya

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EDUCATION

Michigan State University | College of Engineering East Lansing, MI
Bachelor of Science in Electrical Engineering | GPA: 3.7 Expected May 2028
Concentration: Robotics & Smart Devices | **Minor:** Business
Coursework: Electronic Circuits, Circuits & Systems, Digital Logic, Microprocessors & Digital Systems

TECHNICAL SKILLS

Programming: C, C++, Python, Verilog, Java, MATLAB
Hardware & Design: KiCad, Altium, LTspice, Arduino IDE, PCB layout, schematic capture
Tools: Docker, VS Code, Codex, Claude Code, Copilot, Jira, Confluence
Standards: ISO 26262, ISO/SAE 21434, UNECE R155

ENGINEERING EXPERIENCE

Stoneridge Electronics | Novi, MI May 2026 - Present
Functional Safety Engineer (Part-Time); previously Functional Safety Engineering Intern

- Developed **ISO 26262** Item Definitions, **HARA** (Hazard Analysis and Risk Assessment), and Safety Goals for the **MirrorEye Camera Monitoring System** to inform safety requirements and the Functional Safety Concept.
- Led a **Copilot compliance assistant** from development to deployment, integrating standards and enterprise knowledge to cut comparison time **83%** and manual searches **75%**.
- Translated system, functional, and safety requirements into **Hardware Requirements Specifications (HwRS)**, improving hardware requirement quality, consistency, and traceability.
- Assessed customer-driven product, software, and UI changes by tracing requirements, interfaces, verification, and safety artifacts, identifying downstream impacts to inform engineering decisions.
- Modeled system boundaries and dependencies to trace hazards to requirements and safety artifacts; standardized reusable templates to improve documentation consistency across engineering teams.

TriMas Packaging | New Albany, OH Jun 2025 - Aug 2025
Process/Automation Intern

- Cut net rejects by **50%** by diagnosing false suction rejects, standardizing assembly-to-test pushdown distance, and updating **HMI/PLC logic** using station data and part-quality tests.
- Reduced training documentation by **80%** across **4 production lines and 12 machines** by applying Lean/Kaizen to rebuild SOPs and onboarding guides with process engineers.
- Analyzed **2 months of OEE data across 3 shifts** to flag reject anomalies; supported troubleshooting and new product trials through root cause analysis and machine adjustments.

PROJECTS

Autonomous RC Car | Power Distribution PCB Nov 2025 - Present
Electrical Design Lead & Testing Team Member

- Designed a **KiCad 9 power-distribution PCB** integrating a **Raspberry Pi 5**, dual L298N motor drivers, and LiDAR; led schematic capture, PCB layout, and connector integration.
- Stabilized voltage rails during motor transients using split motor/logic grounds, a net tie, and bulk/decoupling capacitors, reducing motor noise and return-current issues.

ESP32 Stepper Motor Control

- Built an **ESP32-based stepper motor controller** mapping keyboard arrow inputs to rotation direction and speed adjustments, enabling interactive control of motor motion.

ADDITIONAL EXPERIENCE

Reflecting Riches LLC | Co-Founder/Owner | Troy, MI Nov 2023 - May 2025

- Launched a journal and **Shopify storefront**; managed fulfillment, inventory, customer support, and organic marketing, using sales and engagement data to guide weekly improvements.